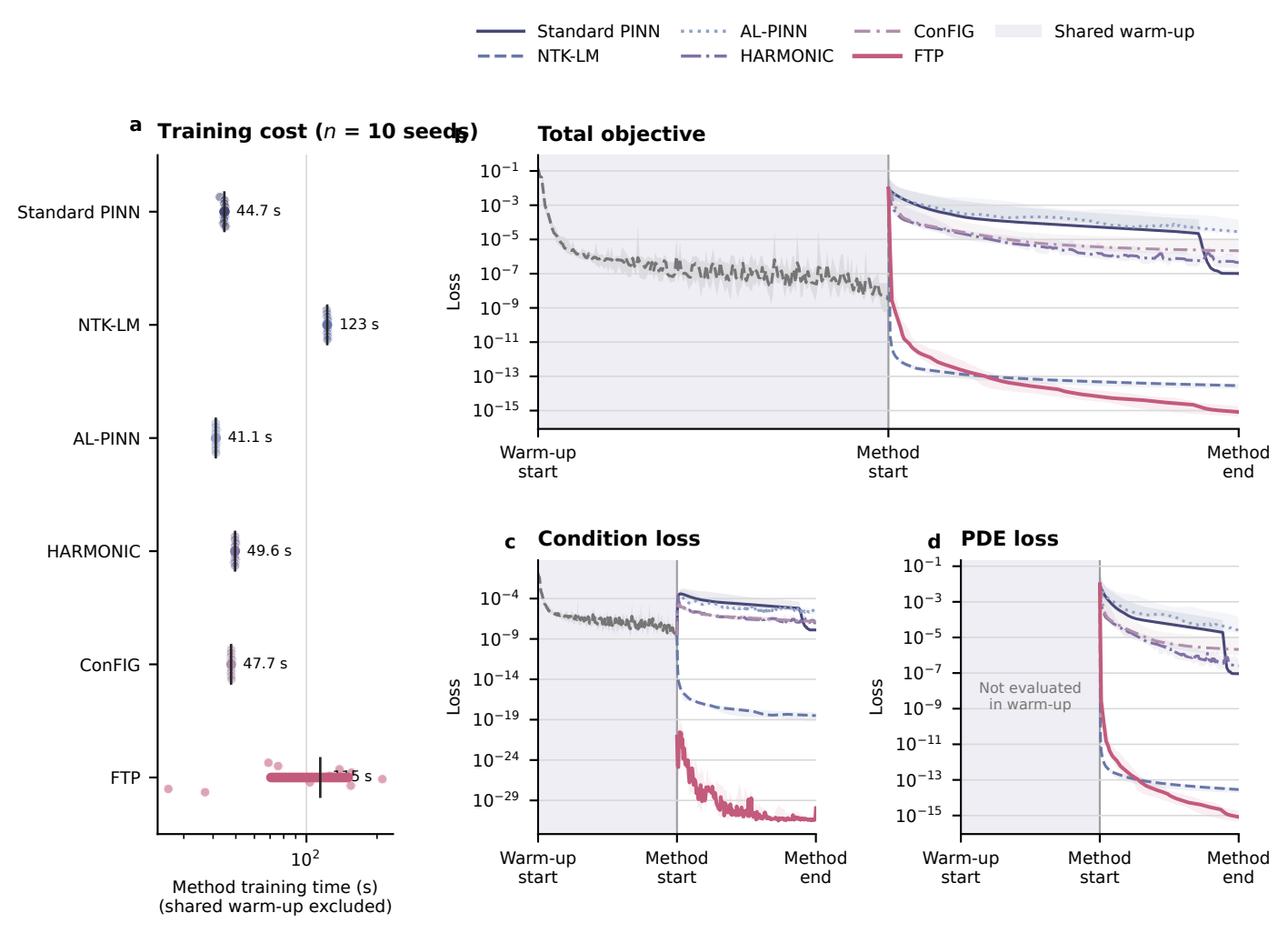

Standard PINN
NTK-LM
AL-PINN
HARMONIC
ConFIG
FTP
Shared warm-up
a
Training cost (n = 10 seeds)
Standard PINN
NTK-LM
AL-PINN
HARMONIC
ConFIG
FTP
44.7 s
123 s
41.1 s
49.6 s
47.7 s
10^2
Method training time (s)
(shared warm-up excluded)
b
Total objective
Loss
Warm-up start
Method start
Method end
c
Condition loss
d
PDE loss
Not evaluated in warm-up